Zhihao Lin

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Education

Santa Clara University Sep 2020 – Dec 2022
Master of Science in Computer Science *Santa Clara, CA*
GPA: 3.77/4.0; GRE: 334

University at Buffalo, The State University of New York Aug 2016 – Jun 2020
Bachelor of Science in Computer Science *Buffalo, NY*
Cum Laude; Dean's List (2019)

Industry Experience

Amazon Web Services — Amazon Connect Feb 2023 – Apr 2026
Software Development Engineer *Seattle, WA*

- Built and operated a large-scale metrics platform for cloud contact center as the service scaled from 10M+ to 20M+ daily customer interactions, supporting the computation, storage, validation, and visualization of product-defined metrics across distributed backend services.
- Owned and productionized an internal metrics validation service using AWS Lambda, DynamoDB, OpenSearch, and S3, expanding pre-launch validation coverage to over 90%, reducing customer complaints related to metric definitions by approximately 70%, and cutting investigation time for related support tickets from one day to two hours.
- Owned CI/CD pipelines, monitoring, and Tier-2 on-call operations for the platform, partnering with PMs and cross-functional engineering teams to launch metrics and resolve production incidents.
- Built retrieval-grounded metric onboarding workflows using DynamoDB and OpenSearch metadata, schemas, dependencies, and calculation logic to automate validation and debugging.
- Developed MCP-style tool-integration infrastructure exposing backend services for schema inspection, metric validation, and script execution, with structured validation guardrails for reliable LLM-driven workflows.
- Introduced the team's first AI-assisted SOP-to-workflow automation prototype, converting operational procedures into structured execution workflows and helping reduce the regional expansion timeline from two weeks to one week.

Amazon Web Services — Amazon Connect Jun 2022 – Sep 2022
Software Development Engineer Intern *Seattle, WA*

- Prototyped an early metrics validation service using AWS Lambda, DynamoDB, OpenSearch, and S3; presented it in an organization-wide demo before subsequently productionizing the service as a full-time engineer.

Research Interests

Adaptive and context-aware LLM generation, representation learning, and AI systems. I am particularly interested in how language models can infer interaction context and adapt generation or tool use accordingly, toward reliable agents that operate across increasingly multimodal and interactive environments.

Publications

Read the Room Before Your Model Generates 2026
Muyao Kong, **Zhihao Lin*** — Under EMNLP 2026 review; ARR meta-review recommendation: Findings*
A scenario-adaptive decoding framework that derives prompt-level sampling temperature from interpretable scenario representations learned from Transformer hidden states. *Equal contribution.

Research Experience

Read the Room Before Your Model Generates Jan 2026 – Present

Independent Researcher

Initial guidance from Prof. Haifeng Xu, University of Chicago; research conducted independently.

- Co-developed Scenario-Actor, a prompt-level decoding-temperature selection framework that infers latent communicative scenarios from Transformer hidden states and maps them to adaptive sampling temperatures
- Implemented and analyzed Scenarize, an RNN-based encoder over Llama-3.1-8B-Instruct hidden states that learns 32-dimensional scenario embeddings using a squared-Euclidean triplet margin loss, and Actout, an interpretable SemAxis-style module that derives decoding temperature from scenario axes.
- Curated and GPT-4o-annotated 37K+ conversational prompts from dialogue corpora across 9 scenario labels; validated annotation quality against human raters using inter-rater agreement analysis.
- Benchmarked Scenario-Actor against greedy decoding, fixed-temperature decoding, nucleus sampling, EDT, and random-temperature baselines on GSM8K, AlpacaEval, and MT-Bench; achieved the highest LC-WR on AlpacaEval among evaluated decoding baselines.
- Conducted layer-sweep and representation-geometry diagnostics, including PCA/UMAP, effective rank, tanh saturation, and scenario-axis off-diagonal correlation analysis, to select the probing layer and verify that the learned scenario space was structured rather than collapsed.
- ARR-reviewed manuscript under consideration at EMNLP 2026.

Sentiment Analysis for Email Interaction Jan 2022 – Apr 2022

Research Assistant, Santa Clara University *Santa Clara, CA*

Advised by Prof. Yi Fang.

- Contributed to an industry-collaborative NLP research project on sentiment analysis by reviewing related literature, participating in research discussions, and deploying experimental models on AWS EC2.
- Built a Chrome/Gmail sentiment-analysis plugin prototype that integrated NLP inference into email workflows for lightweight sentiment interpretation.

Skills

Programming Languages: Python, Java, TypeScript, Bash

Machine Learning Frameworks & Libraries: PyTorch, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, Matplotlib

Systems & Development Tools: Linux, Git, Docker, Jupyter, LaTeX

Cloud Technologies: AWS Lambda, EC2, SageMaker AI, DynamoDB, OpenSearch, S3, SNS, SQS, CloudWatch, CloudFormation